

Logic Gates and Qubits

Problem 29.1

x	y	XOR	(x + y)	(x + y) mod 2
0	0	0	0	0
0	1	1	1	1
1	0	1	1	1
1	1	0	2	0

This shows that XOR is the same as equivalent $(x + y)$ modulo 2.

Problem 29.2

x	y	AND	XOR	NAND	xNOR
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	0	0	1

Problem 29.3

Shirt	Shoes	Service
0	0	0
0	1	0
1	0	0
1	1	1



Problem 29.4

x	x	AND(x, x)	OR(x, x)	XOR(x, x)
0	0	0	0	0
1	1	1	1	0

$$\text{AND}(x, x) = x$$

$$\text{OR}(x, x) = x$$

$$\text{XOR}(x, x) = 0$$

Problem 29.9

Stage 1:

$$\text{XOR}(0, 1) = 1$$

$$\text{OR}(1, 1) = 1$$

$$\text{NAND}(1, 0) = 1$$

Stage 2:

$$\text{AND}(1, 1) = 1$$

$$\text{xNOR}(1, 1) = 1$$

Stage 3:

$$\text{NOR}(1, 1) = 0$$

Output = 0

Problem 29.8

x	x	y	NOT(x)	NAND(x, x)
0	0	0	1	1
0	0	1	1	1
1	1	0	0	0
1	1	1	0	0

NOT(x) = NAND(x, x)

x	y	AND(x, y)	A = NAND(x, y)	B = NAND(x, y)	NAND(A, B)
0	0	0	1	1	0
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	0	0	1

AND(x, y) = NAND(NAND(x, y), NAND(x, y))